be reduced by enclosing the machine in a gas filled steel chamber at very high pressure.

The high voltage produced in this Van de Graaff generator is used to accelerate positive ions (protons and deuterons) for nuclear disintegrations and other applications.

EXAMPLE 1.24

Dielectric strength of air is 3×10^6 V m^{-1}. Suppose the radius of a hollow sphere in the Van de Graff generator is R = 0.5 m, calculate the maximum potential difference created by this Van de Graaff generator.

Solution

The electric field on the surface of the sphere is given by (by Gauss law)

$$E = \frac{1}{4\pi\epsilon_\circ}\frac{Q}{R^2}$$

The potential on the surface of the hollow metallic sphere is given by

$$V = \frac{1}{4\pi\epsilon_\circ}\frac{Q}{R} = ER$$

Since $V_{max} = E_{max}R$

Here $E_{\max} = 3\times10^6\ \mathrm{V m^{-1}}$. So the maximum potential difference created is given by

$$V_{max} = 3 \times 10^6 \times 0.5$$
$$= 1.5 \times 10^6 \text{ V (or) } 1.5 \text{ million volt}$$

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